

Eenheid 2.3

Note Title

2016/03/10

Limietwette
(Sien Clickup)

Stewart bl 95-104
Studiegids bl. 16

Bepaal indien moontlik

$$\begin{aligned} \textcircled{1} \quad \lim_{x \rightarrow 2} (3x - 5 \cos \pi x + 4 \sin \frac{\pi}{x}) & \quad (\text{Direk}) \\ &= 6 - 5 \overset{=1}{\cos 2\pi} + 4 \overset{=1}{\sin \frac{\pi}{2}} \\ &= 6 - 5 + 4 \\ &= 5 \end{aligned}$$

$$\textcircled{2} \quad \lim_{x \rightarrow 3} \frac{3-x}{x^2-9} \quad \left(\frac{0}{0} \right)$$

faktorisiere!

$$= \lim_{x \rightarrow 3} \frac{3-x}{(x-3)(x+3)}$$

$$= \lim_{x \rightarrow 3} \frac{-(\cancel{x-3})}{(\cancel{x-3})(x+3)}$$

$$= -\frac{1}{6}$$

$$\textcircled{3} \quad \lim_{x \rightarrow \frac{\pi}{2}} \sqrt{x} \sin x$$

$$= \lim_{x \rightarrow \frac{\pi}{2}} \sqrt{x} \quad \lim_{x \rightarrow \frac{\pi}{2}} \sin(x)$$

$$= \sqrt{\frac{\pi}{2}} \times 1$$

$$= \sqrt{\frac{\pi}{2}}$$

$$\textcircled{4} \quad \lim_{x \rightarrow 1} \frac{3-x}{x^2-9} = \frac{\lim_{x \rightarrow 1} 3-x}{\lim_{x \rightarrow 1} x^2-9} = \frac{2}{-8} = -\frac{1}{4}$$

X Direkt ??

$$\begin{aligned} \textcircled{5} \quad \lim_{x \rightarrow 0} \frac{x}{\sqrt{x+4} - 2} & \left(\frac{0}{0} \right) \\ & \text{3} \\ & = \lim_{x \rightarrow 0} \frac{x}{\sqrt{x+4} - 2} \times \frac{\sqrt{x+4} + 2}{\sqrt{x+4} + 2} \\ & = \lim_{x \rightarrow 0} \frac{x (\sqrt{x+4} + 2)}{(\sqrt{x+4})^2 - 2^2} \\ & = \lim_{x \rightarrow 0} \frac{x (\sqrt{x+4} + 2)}{(x+4) - 4} \\ & = \lim_{x \rightarrow 0} \frac{x (\sqrt{x+4} + 2)}{x} \end{aligned}$$

Plan ?? $\frac{x \text{ bo}}{x \text{ onder}}$ met
toegevoegde van
 $\sqrt{x+4} - 2$

Dit is

$$\sqrt{x+4} + 2$$

$$= \lim_{x \rightarrow 0} \sqrt{x+4} + 2 \stackrel{\text{direkt}}{=} 4$$

• Knyptangstelling p 101

$$\forall x \text{ naby } a \text{ en } f(x) \leq g(x) \leq h(x)$$

dan is $\lim_{x \rightarrow a} g(x) = L$.

$$\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} h(x) = L$$

Onthou: Metodes om limiete te bepaal:

- Direkte Substitusie

- Faktorisering ($\frac{0}{0}$)

- Halwe limiete

- Limietwette

- Toegevoegde

- art d
• VA

- Knyptangstelling

• Knyptangstelling
meestal vir $\sin x$ en $\cos x$

- $-1 \leq \sin x \leq 1$

- $-1 \leq \cos x \leq 1$

- $-\frac{\pi}{2} < \arctan x < \frac{\pi}{2}$

Voorbeeld: Bepaal $\lim_{x \rightarrow 0^+} \sqrt{x} \sin \frac{1}{x}$

Opmerking: kan nie limietwette gebruik nie
want $\lim_{x \rightarrow 0^+} \sin \frac{1}{x}$ bestaan nie.

Plan: knyptangstelling: $\lim_{x \rightarrow 0^+} \sqrt{x} \sin \frac{1}{x}$

As $x \neq 0$ dan $-1 \leq \sin \frac{1}{x} \leq 1$

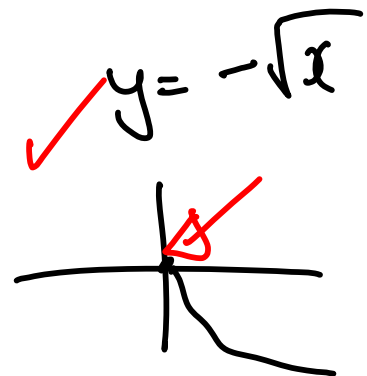
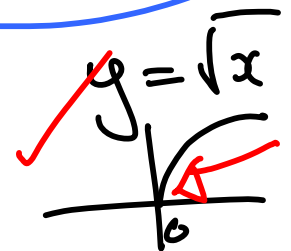
$$\begin{aligned} & \text{(} x\sqrt{x} \text{)} \Rightarrow (-1)\sqrt{x} \leq \sqrt{x} \sin \frac{1}{x} \leq \sqrt{x} (1) \\ \text{Nou: } \lim_{x \rightarrow 0^+} -\sqrt{x} &= 0 \end{aligned}$$

$$\lim_{x \rightarrow 0^+} \sqrt{x} = 0$$

Dan is

(volgens knyptang)

$$\lim_{x \rightarrow 0^+} \sqrt{x} \sin \frac{1}{x} = 0$$



4 Speciale limieten

Dink:
vorm: $\frac{\sin 0}{0}$ (ken dit)
asook toepassings

$$\bullet \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

$$\bullet \lim_{x \rightarrow 0} \frac{x}{\sin x} = 1$$

? Dink
 $\frac{\tan 0}{0}$

$$\bullet \lim_{x \rightarrow 0} \frac{\tan x}{x} = 1$$

$$\bullet \lim_{x \rightarrow 0} \frac{x}{\tan x} = 1$$

VOORBEELDE

$$\textcircled{1} \lim_{x \rightarrow \pi} \frac{\sin x}{x} \stackrel{\text{Direk}}{=} \frac{\sin \pi}{\pi} = \frac{0}{\pi} = 0$$

$$\begin{aligned}
 \textcircled{2} \quad \lim_{x \rightarrow 0} \frac{\sin 4x}{x} & \stackrel{\text{L'H}}{=} \lim_{x \rightarrow 0} \frac{\sin 4x}{\frac{x}{4}} \times \frac{4}{4} \\
 & = 4 \left[\lim_{\substack{x \rightarrow 0 \\ 4x \rightarrow 0}} \frac{\sin 4x}{4x} \right] = 4 \times 1 = 4
 \end{aligned}$$

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

$$\textcircled{3} \quad \lim_{x \rightarrow 2} \frac{\tan(x^2 - 4)}{x - 2} \stackrel{\text{L'H}}{=} ??$$

$$= \lim_{x \rightarrow 2} \frac{\tan(x^2 - 4)}{x - 2} \times \frac{x + 2}{x + 2}$$

$$= \left[\lim_{\substack{x \rightarrow 2 \\ x^2 - 4 \rightarrow 0}} \frac{\tan(x^2 - 4)}{x^2 - 4} \right] \times \lim_{x \rightarrow 2} (x + 2)$$

$$= 1 \times \lim_{x \rightarrow 2} (x + 2) = 1 \times 4 = 4$$

$$\lim_{x \rightarrow 0} \frac{\tan x}{x} = 1$$

④ $\lim_{x \rightarrow 0} \frac{x^2}{\sin x}$ ^{Plan} = $\left[\lim_{x \rightarrow 0} \frac{x}{\sin x} \right] \times \lim_{x \rightarrow 0} x$

$= \lim_{x \rightarrow 0} \frac{x \times x}{\sin x}$

$= 1 \times 0$

$= 0$

$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$

⑤ $\lim_{x \rightarrow 0} \frac{\tan(x^2)}{x}$ ^{LoS} = $\lim_{x \rightarrow 0} \frac{\tan x^2}{x} \times \frac{x^2}{x}$

$= \left[\lim_{x \rightarrow 0} \frac{\tan(x^2)}{x^2} \right] \times \lim_{x \rightarrow 0} x$

$= 1 \times 0$

$= 0$

$$\textcircled{6} \lim_{x \rightarrow 0} \frac{\tan 6x}{\sin 2x} = \lim_{x \rightarrow 0} \tan 6x \times \frac{1}{\sin 2x}$$

$$= \lim_{x \rightarrow 0} \tan 6x \times \frac{6x}{6x} \times \frac{1}{\sin 2x} \times \frac{2x}{2x}$$

$$= \left[\lim_{x \rightarrow 0} \frac{\tan 6x}{6x} \right] \times \cancel{6x} \times \left[\lim_{x \rightarrow 0} \frac{2x}{\sin 2x} \right] \times \cancel{\frac{1}{2x}}$$

$$= 1 \times \lim_{x \rightarrow 0} \cancel{\frac{6x}{2x}} \times 1$$

$$= 1 \times \frac{6}{2} \times 1$$

$$= 3$$

Methodes om limieten te bepaal.

① Direkte substitutie

$$\lim_{x \rightarrow \pi} \frac{\tan x}{x} = \frac{0}{\pi} = 0$$

② Vorm $\left(\frac{0}{0}\right)$: • Factoriseer

Direkt

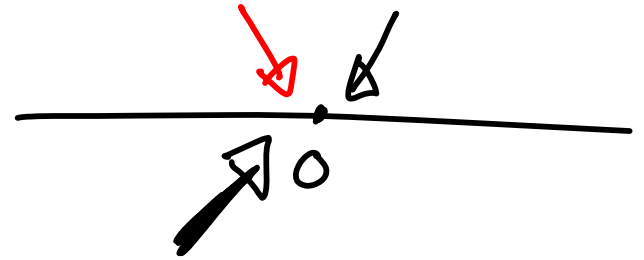
$$\lim_{t \rightarrow -3} \frac{t^2 - 9}{2t^2 + 7t + 3} \left(\frac{0}{0}\right) = \lim_{t \rightarrow -3} \frac{(t-3)(\cancel{t+3})}{(2t+1)(\cancel{t+3})}$$
$$= \frac{-6}{-5}$$
$$= \frac{6}{5}$$

③ Oneindige limiete

$$\begin{aligned} & \lim_{x \rightarrow 0^-} \left(\frac{1}{x} - \frac{1}{|x|} \right) \\ &= \lim_{x \rightarrow 0^-} \left(\frac{1}{x} - \frac{1}{(-x)} \right) \\ &= \lim_{x \rightarrow 0^-} \left(\frac{1}{x} + \frac{1}{x} \right) \\ &= \lim_{x \rightarrow 0^-} \frac{2}{x} = -\infty \end{aligned}$$

Gebruik definisie
van $y = |x|$

$$|x| = \begin{cases} x & \text{as } x \geq 0 \\ -x & \text{as } x < 0 \end{cases}$$



④ Limietwette (op ClickUP)

Gaan kyk

⑤ Toegevoegde

→ onbepaalde vorm

$$\lim_{h \rightarrow 0} \frac{\sqrt{9+h} - 3}{h} \left(\frac{0}{0} \right)$$

$$= \lim_{h \rightarrow 0} \frac{\sqrt{9+h} - 3}{h} \times \frac{\sqrt{9+h} + 3}{\sqrt{9+h} + 3}$$

$$= \lim_{h \rightarrow 0} \frac{9+h-9}{h(\sqrt{9+h}+3)}$$

$$= \lim_{h \rightarrow 0} \frac{\cancel{h}}{\cancel{h}(\sqrt{9+h}+3)}$$

$$= \frac{1}{6}$$

⑥ • Knypstangstelling

Bepaal: $\lim_{x \rightarrow 0} x^4 \cos \frac{2}{x}$

as $x \neq 0$ dan $-1 \leq \cos \frac{2}{x} \leq 1$

moet nou $(x^4 \cos \frac{2}{x})$ skip!

Dan: x^4 dwarsdeur

Dus $(-1)x^4 \leq x^4 \cos \frac{2}{x} \leq (1)x^4$

$$\lim_{x \rightarrow 0} -x^4 = 0 \quad \text{en} \quad \lim_{x \rightarrow 0} x^4 = 0$$

Dan is: $0 \leq \lim_{x \rightarrow 0} x^4 \cos \frac{2}{x} \leq 0$

en $\lim_{x \rightarrow 0} x^4 \cos \frac{2}{x} = 0$ (Knypstangstelling)

⑦ "Vaste Stelling"

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

$$\bullet \lim_{x \rightarrow 2} \frac{\sin(x-2)}{x^2-4} = \lim_{x \rightarrow 2} \frac{\sin(x-2)}{(x-2)} \times \frac{1}{x+2}$$

$$= \left[\lim_{x \rightarrow 2} \frac{\sin(x-2)}{x-2} \right] \times \lim_{x \rightarrow 2} \frac{1}{x+2}$$

$$= 1 \times \frac{1}{4} = \frac{1}{4}$$

Ekstra übde

① $\lim_{x \rightarrow \frac{\pi}{2}} \frac{\cos 2x}{x}$ (Direkt) = $\frac{\cos 2(\frac{\pi}{2})}{\frac{\pi}{2}} = \frac{\cos \pi}{\frac{\pi}{2}} = -\frac{2}{\pi}$

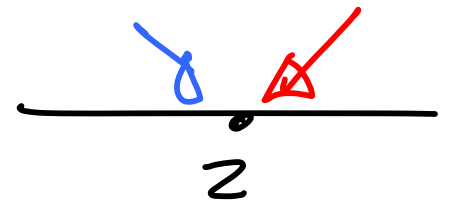
② $\lim_{x \rightarrow 3} \frac{x-3}{x^2-2x-3}$ $\left(\frac{0}{0}\right)$ Faktorisiere = $\lim_{x \rightarrow 3} \frac{(x-3)}{(x-3)(x+1)} = \frac{1}{4}$

kan nie faktorisieren

③ $\lim_{x \rightarrow 2} \frac{(x^2-x+6) \rightarrow 8}{(x-2) \rightarrow 0} = \boxed{?} \infty$

$\lim_{x \rightarrow 2^+} \frac{x^2-x+6 \rightarrow 8}{(x-2) \rightarrow +} = +\infty$

$\lim_{x \rightarrow 2^-} \frac{x^2-x+6 \rightarrow 8}{(x-2) \rightarrow -} = -\infty$

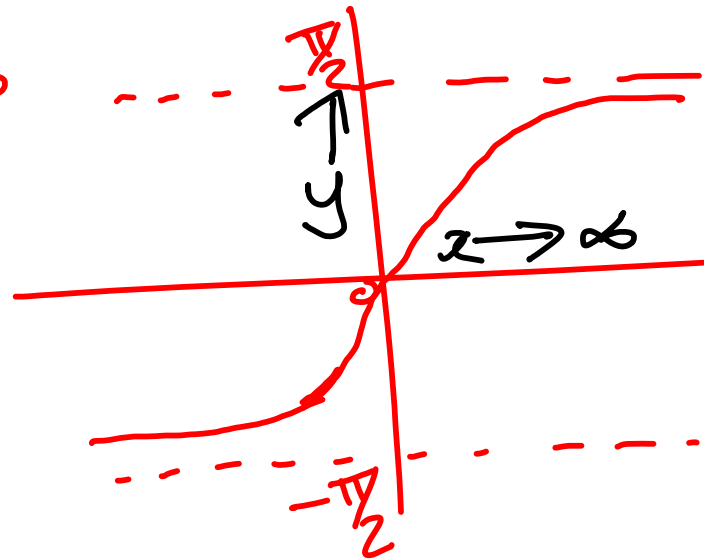


$$\lim_{x \rightarrow 2} \frac{x^2 - x - 6}{x - 2} \text{ bestaan nie}$$

④ $\lim_{x \rightarrow 0^+} \sqrt{x} \arctan \frac{1}{x}$ (2 metodes)

Metode 1: $\lim_{x \rightarrow 0^+} \sqrt{x} \arctan \left(\frac{1}{x} \right)$

$$= 0 \times \frac{\pi}{2} = 0$$



Metode 2: knypvangstelling

As $x \neq 0$ dan $-\frac{\pi}{2} < \arctan \left(\frac{1}{x} \right) < \frac{\pi}{2}$

Skop nou vir $f(x)$: $x \sqrt{x}$

$$\text{Dann } \left(-\frac{\pi}{2}\right)(\sqrt{x}) \leq \sqrt{x} \log \tan \frac{\pi}{2} \leq (\sqrt{x})\left(\frac{\pi}{2}\right)$$

$$\lim_{x \rightarrow 0^+} \left(-\frac{\pi}{2}\right)\sqrt{x} = 0$$

en

$$\lim_{x \rightarrow 0^+} \left(\frac{\pi}{2}\right)\sqrt{x} = 0$$

$$\text{Só dat } \lim_{x \rightarrow 0^+} \sqrt{x} \log \tan \frac{\pi}{2} = 0$$